Badavathu Tejaswini

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.11
KRISHNAVENI JUNIOR COLLEGE	Class XII (TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION)	2017	97.50%
SPARK CONCEPT SCHOOL	Class X (BOARD OF SECONDARY EDUCATION TELANGANA)	2015	9.80%

MASTER THESIS

Digital In-Memory Computing using MAGIC NOR configuration

Objective: To design a Low Latency Multiply and Accumulate(MAC) operation using RRAMs.

Work Done: Optimized a 2-bit multiplier by designing 2-bit carry look ahead adder using RRAM's using Bit-Parallel Element-Parallel Architecture and MAGIC NOR configuration in Cadence Virtuoso. Achieved a low latency of 11 clock cycles compared to 20 cycles in previous work.

Future Work: Now designing a 4-bit carry look ahead adder using the Brent-Kung parallel prefix algorithm to optimize logic depth and delay compared to traditional ripple-carry architecture.

PROJECTS

SRAM-Based 4-Input LUT & CLB Circuit Design

- Designed and simulated a transistor-level 4-input LUT-based Configurable Logic Block (CLB) in LTspice to implement arbitrary combinational and sequential functions.
- Integrated a 16-bit SIPO shift register, 6T SRAM cells, a 4:16 address decoder, and an output D-flip-flop for dual-mode synchronous and asynchronous digital logic operation.

64-Bit Custom SPI Serial-to-Parallel Converter Design:

- Designed and verified a custom 64-bit SPI serial-to-parallel converter featuring hierarchical 8-bit sub-blocks and independent byte-level latch enables using verilog-A.
- Implemented the transistor-level CMOS architecture in Cadence, optimizing D flip-flop, inverter, and transmission gate designs for precise synchronous byte-wise data acquisition.

RTL Debugging & Verification

- Debugged and verified SPUF, OCSPUF, and SPMPUF architectures at RTL level using Verilog and Xilinx Vivado simulation environments.
- Analyzed Arbiter PUF path-delay behavior, identifying timing issues between top and bottom paths that caused stuck response bits.
- Debugged OCSPUF and SPMPUF challenge-response behavior by introducing customized delay offsets across SPUF instances and modifying load time conditions to generate differentiated response patterns.

5T OTA

- Designed and simulated a two-stage CMOS OTA using GPDK 90 nm technology in Cadence Virtuoso, incorporating a differential pair, current-mirror biasing, active load, and Miller compensation.
- Designed the OTA with a 2 V supply, 3 µA bias current, 26 fF compensation capacitor, and 20 fF load capacitor, achieving approximately 68.42 dB nominal gain.

- Performed DC, AC, and PVT analysis across 40°C to 125°C, achieving a maximum gain of approximately 70 dB with stable operation across process corners

SKILLS

Languages: HDL, Verilog, C.

Skills: RTL Design, FSM Design, Digital Logic Design, CMOS Design.

Verification: Testbench Development, Functional Verification, Simulation & Debugging.

Tools: Xilinx Vivado, Cadence Virtuoso, LtSpice, EDA playground.

RELEVANT COURSES

Digital IC design, Introduction to HDL, Memory Device Technology for AI/ML, Digital IC design Lab, CMOS Digital Design, Analog IC Design, Analog IC Design Lab, VLSI Technology, Nanoelectronics: Principles & Devices, Mesoscopic Device Electronics, Semiconductor device modeling.

POSITIONS OF RESPONSIBILITY

- Teaching assistant for Semiconductor Device Fundamentals Dr.Oves Badami and Dr. Shubhadeep Bhattacharjee
- Teaching assistant for Probability and Random Variables by Dr. Anjana A M

WORK EXPERIENCE

Azure Solution Specialist | G7CR Technologies Pvt. Ltd.

- Front-faced startups under the Microsoft for Startups Founders Hub program to drive Azure cloud architecture, optimize customer footprints and spend, capture upsell opportunities, and execute targeted GTM growth strategies.
- Acted as a Customer Success Manager, coordinating across internal technical, finance, and Microsoft program management teams to ensure seamless operations and maximize customer satisfaction.
- Partnered directly with startup stakeholders, Microsoft Technical Account Managers, and Program Managers to deliver end-to-end cloud migration, onboarding, and ongoing solution support.
- Mentored and onboarded 3 team members, delivering comprehensive training on customer success workflows, Azure solutioning, and account management practices.